

When Protection Works but the Portfolio Still Lags

Long-Horizon Evidence on Downside Protection, Benchmark-Relative Drawdowns, and Persistent Equity Exposure

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Abstract

A defensive strategy can meet its downside objective and still lag equity for years. The analysis evaluates both dimensions within the same rolling holding periods. The design is intentionally simple: U.S. equity buy-and-hold, a monthly rebalanced 50/50 equity-cash allocation, and a 12-month defensive trend rule whose position uses only information available through the previous month. The evaluation covers 5-, 10-, 15-, and 20-year periods, treating the signal horizon and blend weights as fixed design choices rather than tuning them to maximize the reported results. In the July 1927-June 2026 French-series sample, the trend rule reduces maximum drawdown in 98.2% of 20-year periods. Among those protected periods, however, 60.3% finish with lower terminal wealth than equity buy-and-hold, and the median longest benchmark-relative underwater period lasts 195 months. Fixed combinations of persistent equity exposure and tactical defense reveal a second distinction: greater persistent equity exposure is associated with substantially shallower benchmark-relative drawdowns, while underwater duration changes much less. The qualitative pattern remains under fixed trading-cost assumptions and in an independent reconstruction using Shiller and FRED data, although the magnitudes differ. The analysis considers realized downside protection together with the benchmark-relative path experienced while obtaining it. These are distinct portfolio-design dimensions and should be assessed separately.

1. Two Dimensions of Defensive Performance

A defensive portfolio can meet its stated downside objective and still spend years lagging its benchmark.

Consider an allocation designed to reduce large equity losses. If a severe bear market arrives and the strategy loses materially less than equity, it has met that objective in that episode. Yet the same allocation may have trailed equity for years before protection was needed and may lag again after risk assets recover. These outcomes raise two distinct questions:

1. Did the strategy reduce drawdown?
2. What benchmark-relative performance path accompanied that protection?

The first concerns drawdown in the strategy's own wealth path. The second concerns performance relative to the benchmark. They need not give the same answer.

The underlying concepts are well established. Mantilla-García (2014) develops absolute and benchmark-relative drawdown-control objectives, while Mahmoud (2017) formalizes drawdown duration as a temporal dimension of path-dependent risk. Sekine and Wunsch (2026) study persistent benchmark-relative underperformance as a portfolio-control problem. Related practitioner research discusses the governance difficulty created by prolonged strategy underperformance (Clements, West, and Kalesnik, 2017), while Hawkins and Gambhir (2024) examine minimum holding periods for benchmark outperformance.

This paper asks a narrower historical question: among holding periods in which a strategy actually reduced maximum drawdown, what benchmark-relative path accompanied that protection? The analysis reports

relative drawdown depth, underwater duration, and terminal benchmark outcome separately rather than compressing them into a single score.

That separation matters. A holding period in which the strategy does not reduce maximum drawdown provides no realized downside protection under this definition. A period in which it does reduce drawdown can still end with benchmark underperformance. Recording the two outcomes separately avoids treating one as evidence for or against the other.

2. A Deliberately Simple Empirical Design

The timing rule is intentionally simple because the objective is evaluation rather than signal discovery.

The long-history analysis uses monthly data from the Kenneth R. French Data Library. Monthly market total return is the market excess return plus the one-month Treasury bill return. The source series runs from July 1926 through June 2026; the first trend position is formed for July 1927 after 12 completed months of return history. French's documentation defines the market excess return as the value-weighted return on eligible U.S. CRSP firms minus the one-month Treasury bill rate.

The analysis compares three transparent allocations:

- Equity buy-and-hold: the benchmark.
- 50/50 equity-cash: rebalanced monthly, as a simple non-timing defensive comparison.
- 12-month defensive trend: invested in equity for month t when compounded market return over the prior 12 completed months exceeds compounded cash return over the same window, and otherwise invested in cash.

The position for month t therefore uses information only through month $t-1$.

Each month starts a new hypothetical holding period. The evaluation follows each rolling period for 5, 10, 15, or 20 years whenever the data permit. Because these periods overlap, the reported frequencies summarize the historical sample rather than outcomes from independent observations.

The 12-month signal horizon and the blend weights are treated as fixed design choices rather than tuned to maximize the reported results. The blend analysis reports tactical shares of 0%, 25%, 50%, 75%, and 100% to characterize how the measured trade-offs vary across fixed allocations.

For each rolling period, maximum drawdown (MDD) is the largest peak-to-trough loss. Four measures are then recorded:

- MDD protection: the percentage-point reduction in maximum-drawdown magnitude relative to equity;
- terminal benchmark outcome: whether terminal wealth finishes above or below equity buy-and-hold;
- benchmark-relative drawdown depth: the largest decline in the strategy-to-equity wealth ratio from a prior relative high-water mark;
- benchmark-relative underwater duration: the longest spell below a prior relative high-water mark.

Depth and duration remain separate. A portfolio change can improve one without materially improving the other.

Long-history evidence documents trend-following behavior across markets and crises (Hurst, Ooi, and Pedersen, 2017). Recent work also compares defensive strategies over more than two centuries of global history (Baltussen, Martens, and van der Linden, 2026). Here, a single transparent trend rule serves as an empirical vehicle for studying how defensive portfolios should be evaluated.

3. Protection and Benchmark Outperformance Are Different Events

Figure 1 summarizes the central descriptive result.

At the 5-year horizon, the trend rule reduces MDD in 74.4% of rolling periods. The frequency rises to 90.0% at 10 years, 98.3% at 15 years, and 98.2% at 20 years. In this sample, longer holding periods are more often protected, consistent with a greater chance of spanning a material equity drawdown.

At 20 years, protection is common but benchmark outperformance is not. Of all 20-year periods, 59.2% are in the protected + underperformed state. Among protected periods alone, 60.3% still finish with less wealth than equity buy-and-hold.

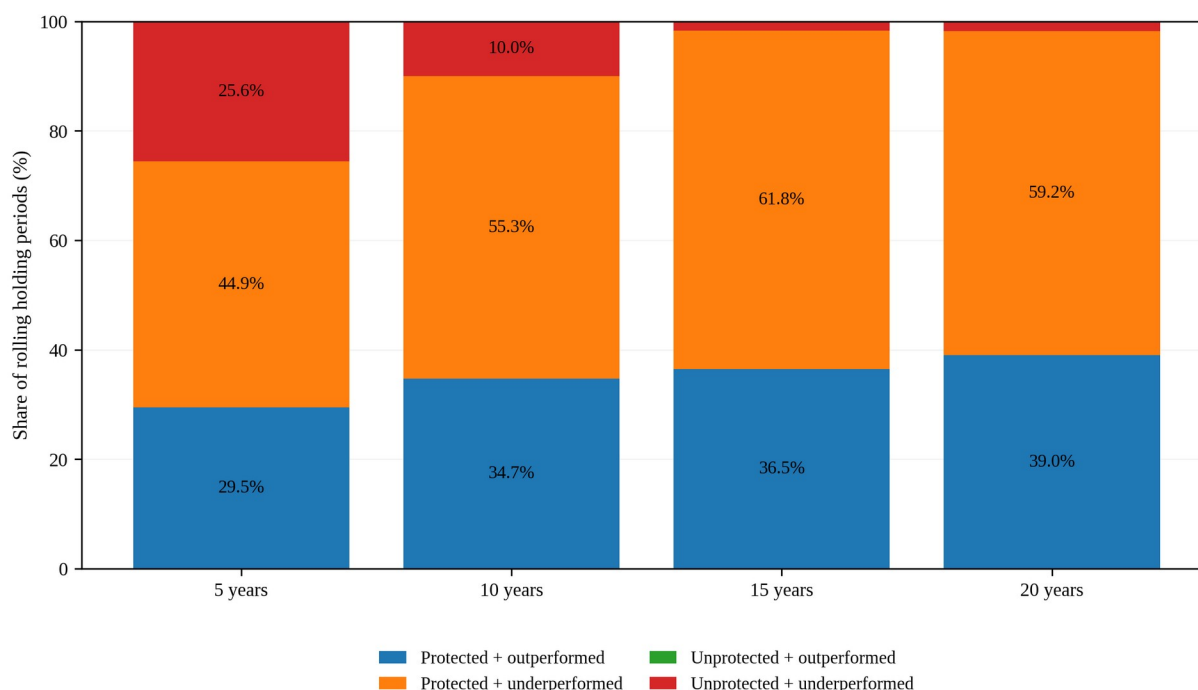


Figure 1. Downside protection and benchmark outperformance are different outcomes. Four-state decomposition of rolling holding periods for the 12-month defensive trend rule versus equity buy-and-hold. Periods overlap; frequencies are descriptive. No unprotected-outperforming periods appear at the four fixed horizons.

This is the central tension. In most long-horizon periods, the strategy records a lower maximum drawdown than equity. Yet a majority of those protected periods still finish behind the benchmark.

The same distinction appears without tactical timing. The monthly rebalanced 50/50 equity-cash allocation reduces MDD in every 20-year period in the French-series sample but finishes below equity in 98.8% of them. The coexistence of drawdown protection in the strategy's own wealth path and relative underperformance is therefore not unique to the trend rule.

Figure 1 also distinguishes outcomes that a binary classification would obscure. Four states are conceptually possible: protected and outperforming, protected and underperforming, unprotected and outperforming, and unprotected and underperforming. No unprotected-outperforming periods occur at the four fixed horizons in this sample; that absence is a historical result, not a general property of defensive trend strategies.

4. Benchmark-Relative Shortfall Has Both Depth and Duration

Terminal wealth alone does not describe the path of relative underperformance.

Among protected periods, median benchmark-relative drawdown deepens with horizon: 22.6%, 30.9%, 39.5%, and 44.4% at 5, 10, 15, and 20 years, respectively. These are not portfolio losses. They are declines from a prior high-water mark in the strategy's wealth relative to equity.

Duration is equally revealing. The median longest benchmark-relative underwater spell among protected periods is 39 months at 5 years, 89 months at 10 years, 144 months at 15 years, and 195 months at 20 years. The 20-year median is 195 months, or 16.25 years.

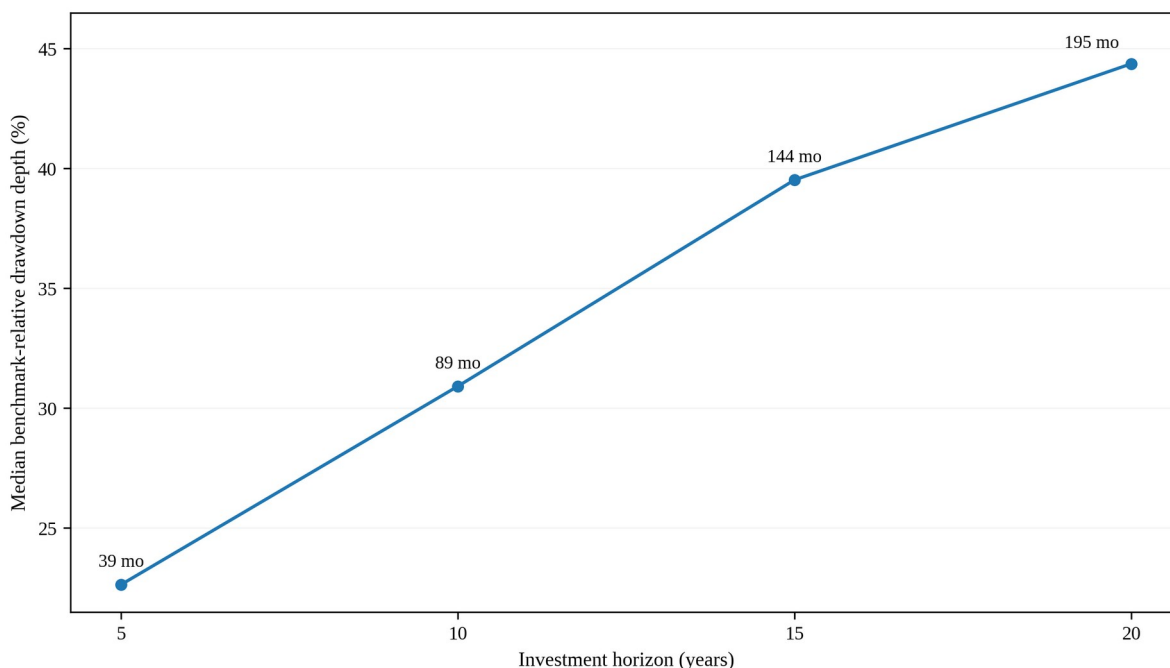


Figure 2. Protected holding periods can remain below their prior relative high-water mark for years. The vertical axis is benchmark-relative drawdown depth among protected holding periods; labels report the median longest benchmark-relative underwater duration in months.

A strategy can therefore reduce its worst peak-to-trough drawdown while spending much of a long holding period below a prior benchmark-relative high-water mark. This is distinct from volatility, drawdown in the strategy's own wealth path, and terminal benchmark underperformance.

Aggregate medians can still make that experience feel abstract. Figure 3 therefore shows one recent 20-year path selected by a simple reproducible rule: among 20-year periods in which the trend rule both reduces maximum drawdown and finishes below equity, choose the latest window whose longest benchmark-relative underwater spell is closest to the 195-month median reported for all protected 20-year periods. This selects June 2005-May 2025. The example is illustrative rather than an additional statistical test.

Over this window, equity experiences a 50.3% maximum drawdown, while the trend rule's maximum drawdown is 23.2%, a 27.1 percentage-point reduction. Yet the trend strategy finishes with only 70.4% of the benchmark's terminal wealth, and its longest benchmark-relative underwater spell lasts 195 months. The path makes the timing asymmetry tangible: protection is earned in relatively concentrated stress episodes, while benchmark-relative opportunity cost can accumulate across much longer intervals.

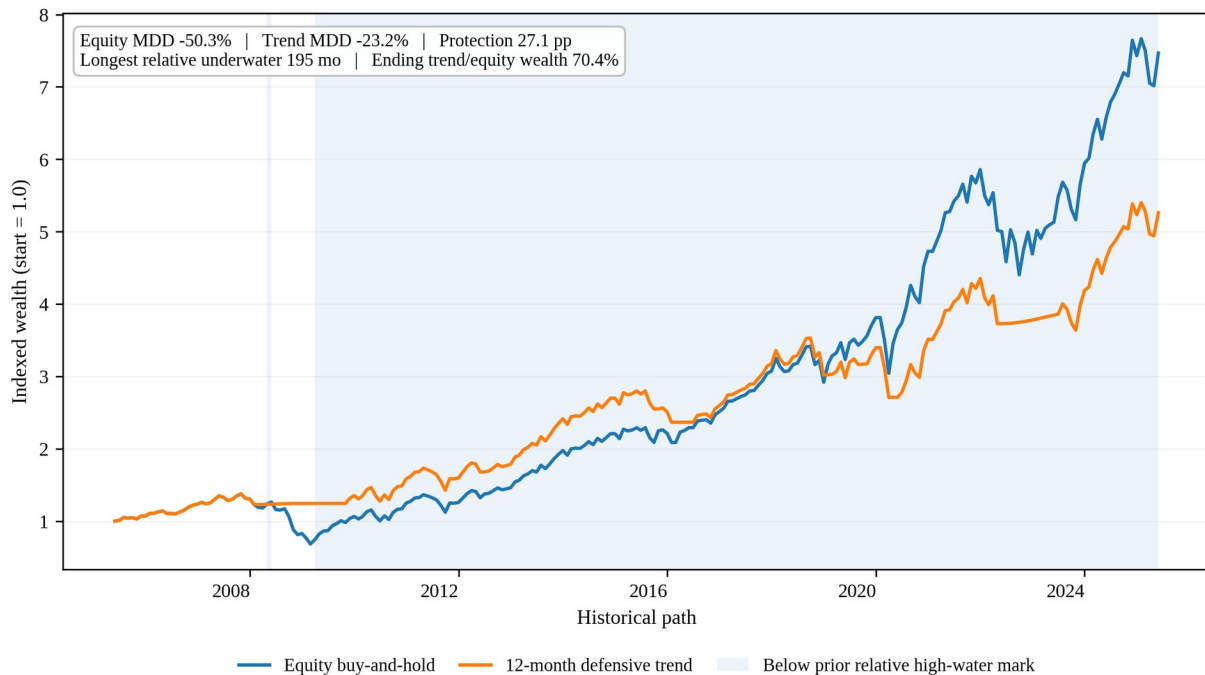


Figure 3. A protected portfolio can still spend most of a 20-year window below its benchmark-relative high-water mark. Indexed wealth for equity and the 12-month defensive trend rule from June 2005 through May 2025. Shading marks months in which the trend-to-equity wealth ratio is below its prior relative high-water mark. The path is illustrative and selected by the reproducible rule stated in the text.

These results do not imply a mechanical trade-off in which greater protection necessarily causes a larger benchmark-relative shortfall. During a major bear market, defensive positioning can reduce the strategy's own drawdown and outperform equity at the same time. The relative benefit of defensive positioning can be concentrated during sharp equity declines, whereas benchmark-relative performance accumulates over the entire holding period.

Defensive gains are also concentrated in time. Using a fixed criterion to identify months of defensive outperformance - a negative equity month in which the strategy exceeds equity by at least 2 percentage points - the trend rule records 112 such months. The 20 largest account for 36.2% of total positive monthly active log returns after the 12-month signal becomes available; the static 50/50 allocation accounts for 19.6%. The median gap between qualifying months is two months. Because adjacent crisis months can belong to the same market episode, these event counts are interpreted descriptively. The broader point is that defensive gains can arrive in concentrated bursts while benchmark-relative performance is observed continuously.

5. Persistent Equity Exposure Changes the Relative Path

This distinction shifts the question from choosing a stand-alone strategy to constructing the overall portfolio.

A natural portfolio-level question is whether retaining equity exposure alongside the tactical strategy changes the benchmark-relative path while preserving some drawdown protection. Fixed combinations of buy-and-hold and tactical exposure already appear in practitioner research. Counterpoint Funds (2018), for example, illustrates a 50/50 blend. The question here is narrower: across fixed weights, what happens separately to benchmark-relative depth and duration as persistent equity exposure increases?

Figure 4 shows the 20-year result across five fixed tactical shares. Pure tactical exposure produces a median reduction in MDD of 22.9 percentage points and a median benchmark-relative drawdown depth of 44.4%. At 75% tactical, the median MDD reduction remains 21.2 percentage points while relative depth falls to 35.1%.

With tactical shares of 50% and 25%, the median reductions in MDD are 18.0 and 8.1 percentage points, and relative depth falls to 24.6% and 12.9%, respectively.

The fixed weights are used to illustrate how benchmark-relative depth and duration change as persistent equity exposure increases, rather than to estimate an optimal allocation.

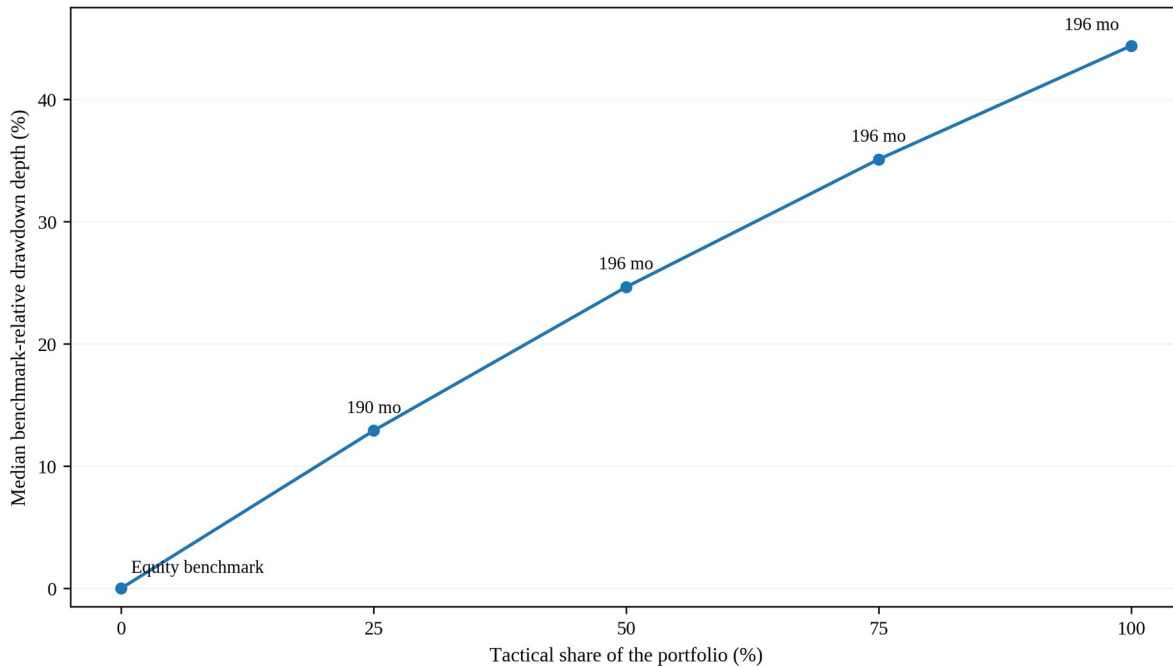


Figure 4. Persistent equity exposure reduces benchmark-relative drawdown depth much more than duration. Five fixed tactical shares over 20-year holding periods; labels report median longest benchmark-relative underwater duration.

Relative drawdown depth changes materially as persistent equity exposure is added. Underwater duration changes much less. Across all 20-year holding periods in the French-series analysis, the median longest benchmark-relative underwater period is 196 months for pure tactical exposure. It is also 196 months at 75% and 50% tactical, and 190 months at 25% tactical. Figure 4 uses all 20-year holding periods, whereas Figure 2 conditions on periods in which the tactical strategy reduced maximum drawdown. This difference in conditioning accounts for the 196-month and 195-month medians reported for pure tactical exposure.

Across these fixed blends, greater persistent equity exposure is associated with substantially shallower benchmark-relative drawdowns, while median underwater duration changes much less.

This distinction matters in practice. A portfolio need not eliminate a prolonged relative shortfall to reduce its severity. Return data cannot identify the point at which an investor, committee, or client would abandon a strategy, so no such threshold is estimated. Evidence from delegated asset management shows that prior performance is associated with fund flows and manager changes, making the path of relative performance relevant to institutional decisions (Bessler et al., 2018).

These results show that stand-alone protection statistics are insufficient to characterize the investor's benchmark-relative path. Portfolio-level allocation affects the depth and duration of benchmark-relative shortfalls associated with holding the defensive component.

6. Robustness to Data Construction and Trading Costs

A historical analysis spanning roughly a century can still depend materially on one data construction. To assess sensitivity to data construction, the same analysis is repeated using an independent reconstruction.

The independent equity series is reconstructed from Robert Shiller's monthly U.S. price and dividend data. It is not treated as an exact tradable-index replication because parts of the long history use monthly average prices and interpolated dividends. Cash is reconstructed from the Federal Reserve's monthly 3-month Treasury bill secondary-market discount-basis series from FRED, using the previous month's rate and a fixed 91-day-bill conversion. This reconstruction starts later because the Treasury bill series begins in the 1930s.

For comparability, both constructions are evaluated over the common February 1935-June 2023 sample. The correlation of monthly equity returns is about 0.66, while the correlation of calendar-year returns is about 0.98, and the 12-month trend signal agrees in about 95.9% of months. The lower monthly correlation shows that the two return series differ materially at the monthly frequency, while the high annual correlation and signal agreement show close alignment in broader annual movements and signal classifications.

At the 20-year horizon, the trend rule reduces MDD in 97.9% of rolling periods in the matched French series and 91.1% in the Shiller/FRED reconstruction. Median MDD protection is 22.3 versus 14.2 percentage points. The benchmark-relative path remains difficult in both: median relative drawdown depth is 43.2% versus 38.5%, and median longest underwater duration is 197 versus 189 months.

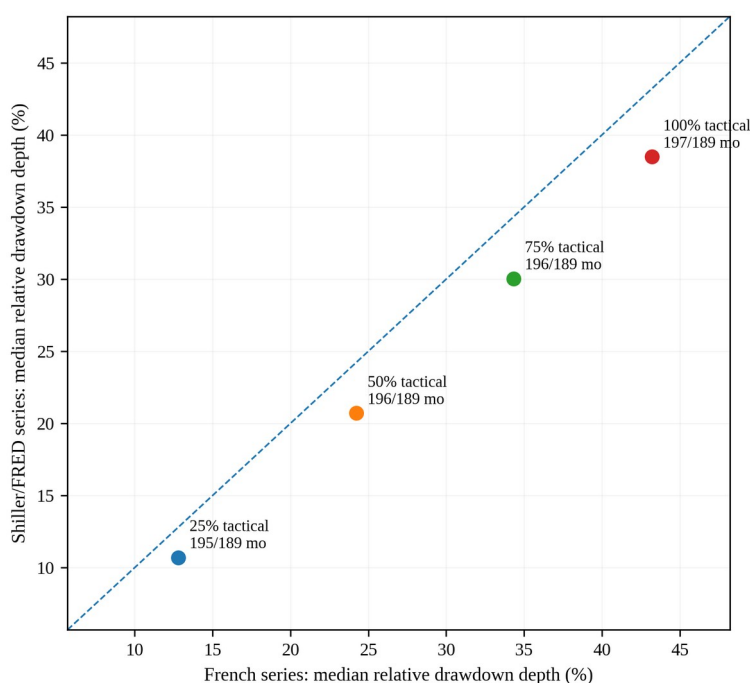


Figure 5. The main pattern also appears in an independent historical reconstruction. Matched February 1935-June 2023 French versus Shiller/FRED comparison; labels report median underwater duration for the French and Shiller/FRED constructions, respectively. The dashed line is the 45-degree reference.

The same depth-versus-duration pattern appears in both constructions. Lower tactical shares materially reduce benchmark-relative drawdown depth, while median underwater duration changes much less. The magnitudes differ, so the independent reconstruction supports the direction and economic interpretation of the main finding rather than reproducing the same numerical magnitudes.

The same distinction remains visible under fixed trading costs. One-way costs of 0, 10, 25, and 50 basis points are applied to monthly portfolio turnover, with no inception charge. At 20 years, the trend rule still reduces MDD in about 98.2% of rolling periods under cost assumptions up to 25 basis points and in about 97.7% at 50 basis points. Trading costs worsen benchmark-relative performance but do not remove the coexistence of downside protection and a long relative shortfall.

These robustness comparisons do not establish that one simple trend rule is universally attractive. They show that the distinction between protection against drawdown in the strategy's own wealth path and benchmark-relative shortfall remains visible under the alternative reconstruction and the fixed trading-cost assumptions.

7. Implications for Portfolio Design

The evidence has four practical implications for evaluating and sizing defensive allocations.

Define the defensive objective first

If downside reduction is the stated objective, evaluation should begin with whether the allocation reduces drawdown in its own wealth path. Benchmark-relative return remains important, but it is a separate dimension rather than a substitute for the stated defensive objective. The two dimensions should therefore be evaluated separately.

Measure the path as well as the endpoint

A long-run CAGR or terminal wealth comparison cannot reveal whether a strategy spent a decade or more below a prior relative high-water mark. When a defensive allocation is judged against an equity benchmark, relative depth and duration should be evaluated alongside drawdown protection in the strategy's own wealth path.

Set benchmark-relative tolerance before implementation

A long period of benchmark underperformance is a poor time to discover that the portfolio was sized beyond the investor's tolerance. Benchmark-relative tolerance is therefore an implementation constraint that should be considered when sizing a defensive allocation. Historical paths can inform that assessment, although they do not determine an investor-specific threshold.

Use persistent equity exposure deliberately

Persistent equity exposure is commonly treated as a return source. Here it also changes the benchmark-relative path. Across the fixed blends, greater persistent equity exposure is associated with substantially shallower benchmark-relative drawdowns, while the longest underwater period changes much less.

This gives blending an additional role. Beyond combining return sources or reducing volatility, an always-invested equity component can preserve some drawdown protection while reducing the severity of benchmark-relative drawdowns.

These comparisons characterize the trade-off across fixed blends rather than identify an optimal tactical weight. The portfolio question is whether a chosen mix is consistent with both the desired level of downside protection and a benchmark-relative path the investor is willing and able to maintain.

8. Scope and Limitations

Several limitations define the scope of the evidence:

- U.S. historical scope. The analysis uses U.S. return paths and does not establish that the same magnitudes hold in every country, asset class, or future regime.
- Overlapping holding periods. Because the rolling holding periods overlap, they are strongly dependent. The reported frequencies therefore summarize the historical sample and should not be interpreted as probabilities estimated from independent trials.
- Investor behavior. Return data do not identify an investor's actual abandonment threshold. Statements about client tolerance or manager retention are practical implications informed by external literature, not behaviors estimated from these returns.
- Independent reconstruction. The Shiller/FRED reconstruction uses a different historical data construction from the French/CRSP market series. Its purpose is to assess whether the qualitative pattern remains visible rather than to reproduce the same numerical results.

- **Methodological scope.** The analysis is empirical rather than methodological. It applies established drawdown and benchmark-relative measures to the same rolling holding periods and examines how their joint behavior changes across fixed portfolio blends over long histories. The timing rule serves as a transparent empirical example rather than an estimated optimal allocation.

9. Conclusion

The benefits and costs of defensive investing become visible at different times. Downside protection matters most during severe losses, while the opportunity cost relative to equity can accumulate during the much longer intervals between them.

That asymmetry creates a practical problem. A strategy may reduce drawdown during severe equity losses and still remain below a prior benchmark-relative high-water mark for prolonged periods. Adding persistent equity exposure can make that shortfall materially less severe without making it much shorter.

The evidence therefore points to two portfolio-design objectives that should be evaluated separately: the degree of drawdown protection sought in the portfolio's own wealth path, and the benchmark-relative path the investor can realistically sustain.

For portfolio design, this distinction leads to two separate questions. How much drawdown protection is required? What benchmark-relative path can the investor sustain long enough for that protection to remain in place when it is needed?

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Data and code availability

Public replication materials are available at <https://github.com/slackquant80/protection-patience-replication>. The repository includes the research code, public source metadata and paper-snapshot hashes, author-generated reference tables, publication figures, and reproducibility checks used for this working paper. Downloaded Kenneth R. French, Shiller, and FRED source files are not redistributed; the repository provides source metadata and retrieval instructions for those public inputs.

Research Platform

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OpenAI's ChatGPT was used for limited technical and editorial assistance during manuscript preparation. The author reviewed and verified the analyses and text and takes full responsibility for the content.